

Sportswear Inventory Manager

n8n Inventory Management Workflow - Assignment Report

Built with n8n, Google Sheets, Groq AI Agent and Hoppscotch

INTER AI Club UMT

Submitted By: Daniyal Arqam Talha

Submitted To: Sir Umair Khan

Assignment: (#8) N8n

1. Project Overview

For this assignment, I created a Sportswear Inventory Manager in n8n. The workflow receives product data through a webhook, checks whether the data is valid, detects duplicate SKUs, stores valid products in Google Sheets, and uses an AI Agent when an item is running low on stock. I kept the normal validation steps rule-based so the results stay predictable, while the AI Agent is used for the part where a recommendation is useful.

2. Inventory Data Model

The main inventory record contains Item Name, SKU, Category, Size, Price, Stock Quantity, Supplier, Stock Status and Time Stamp. Size is treated as optional; when it is not provided, the workflow stores N/A.

3. Workflow Design

The workflow starts with a POST webhook. The first IF node validates the incoming data. Invalid data is returned immediately with a 400 response. Valid data goes to a low-stock check. If stock is 5 or below, the AI Agent is used before the product continues through the normal inventory process. The workflow then checks Google Sheets for an existing SKU. A duplicate returns 409 Conflict, while a new item is appended to the main inventory sheet and returns 201 Created.

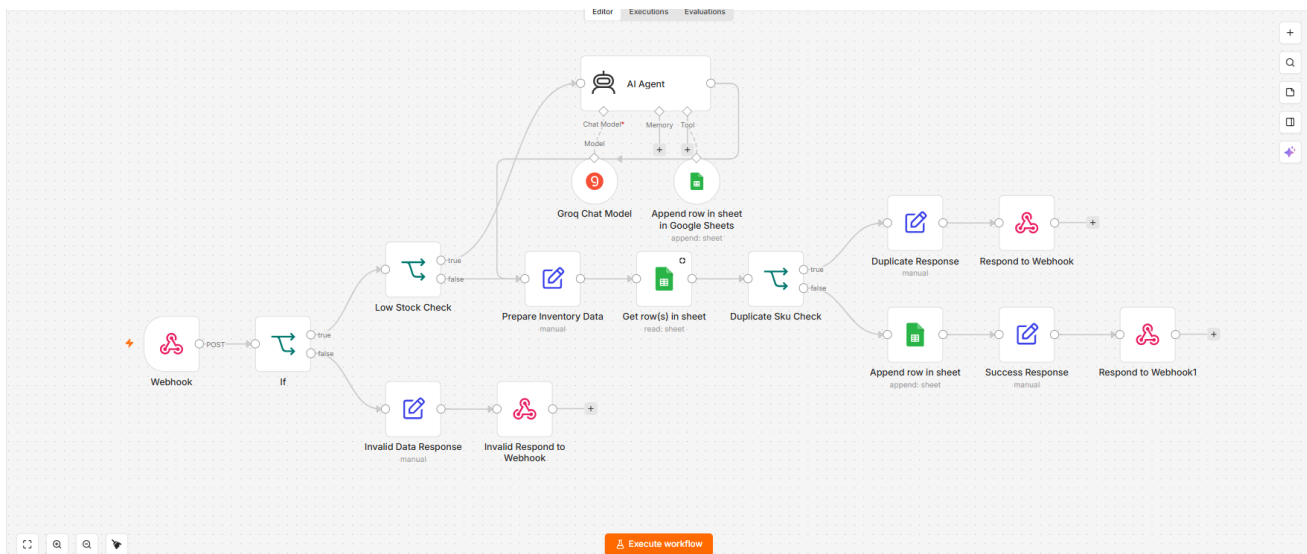


Figure 1: Complete Sportswear Inventory Manager workflow in n8n.

4. Validation and Edge Cases

The workflow checks that item name, SKU, category and supplier are present. Price must be greater than zero and stock quantity cannot be negative. These checks stop bad inventory records before they reach Google Sheets. I also tested optional data and duplicate SKUs instead of testing only the happy path.

Test	What was checked	Result
Valid product	A normal product with stock 20 was accepted.	201 Created
Negative stock	A product with stock quantity -4 was rejected.	400 Bad Request
Zero price	A product with price 0 was rejected after changing the price rule to require a value greater than zero.	400 Bad Request
Missing SKU	An empty required SKU was rejected.	400 Bad Request
Missing optional size	A product without a size was accepted and stored with N/A.	201 Created
Duplicate SKU	A second request using TEST-VALID-001 was blocked.	409 Conflict

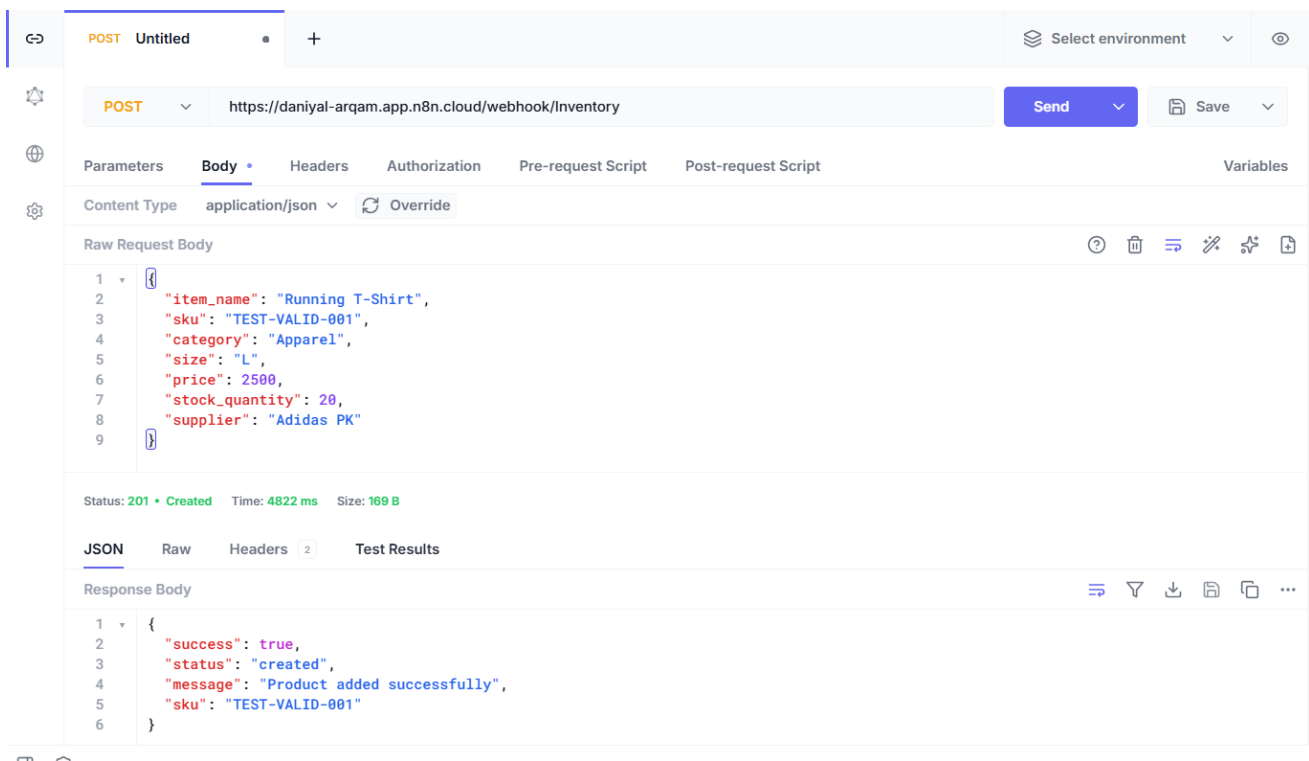


Figure 2: Valid product request successfully returned 201 Created.

POST Untitled • POST Untitled • POST Untitled • POST Untitled • + Select environment

POST https://daniyal-arqam.app.n8n.cloud/webhook/Inventory Send Save

Parameters Body Headers Authorization Pre-request Script Post-request Script Variables

Content Type application/json Override

Raw Request Body

```
1 {
2   "item_name": "Sports Cap",
3   "sku": "TEST-NEG-003",
4   "category": "Accessories",
5   "size": "One Size",
6   "price": 1200,
7   "stock_quantity": -4,
8   "supplier": "Puma PK"
9 }
```

Status: 400 • Bad Request Time: 866 ms Size: 153 B

JSON Raw Headers 2 Test Results

Response Body

```
1 {
2   "success": false,
3   "status": "validation_failed",
4   "message": "Invalid inventory data"
5 }
```

Figure 3: Negative stock quantity correctly failed validation.

HOPPSCOTCH Search and commands Ctrl K

POST Untitled • POST Untitled • POST Untitled • POST Untitled • + Select environment

POST https://daniyal-arqam.app.n8n.cloud/webhook/Inventory Send Save

Parameters Body Headers Authorization Pre-request Script Post-request Script Variables

Content Type application/json Override

Raw Request Body

```
1 {
2   "item_name": "Gym Bottle",
3   "sku": "TEST-PRICE-004",
4   "category": "Accessories",
5   "size": "One Size",
6   "price": 0,
7   "stock_quantity": 15,
8   "supplier": "Nike PK"
9 }
```

Status: 400 • Bad Request Time: 1295 ms Size: 153 B

JSON Raw Headers 2 Test Results

Response Body

```
1 {
2   "success": false,
3   "status": "validation_failed",
4   "message": "Invalid inventory data"
5 }
```

Figure 4: Zero-price product correctly failed validation.

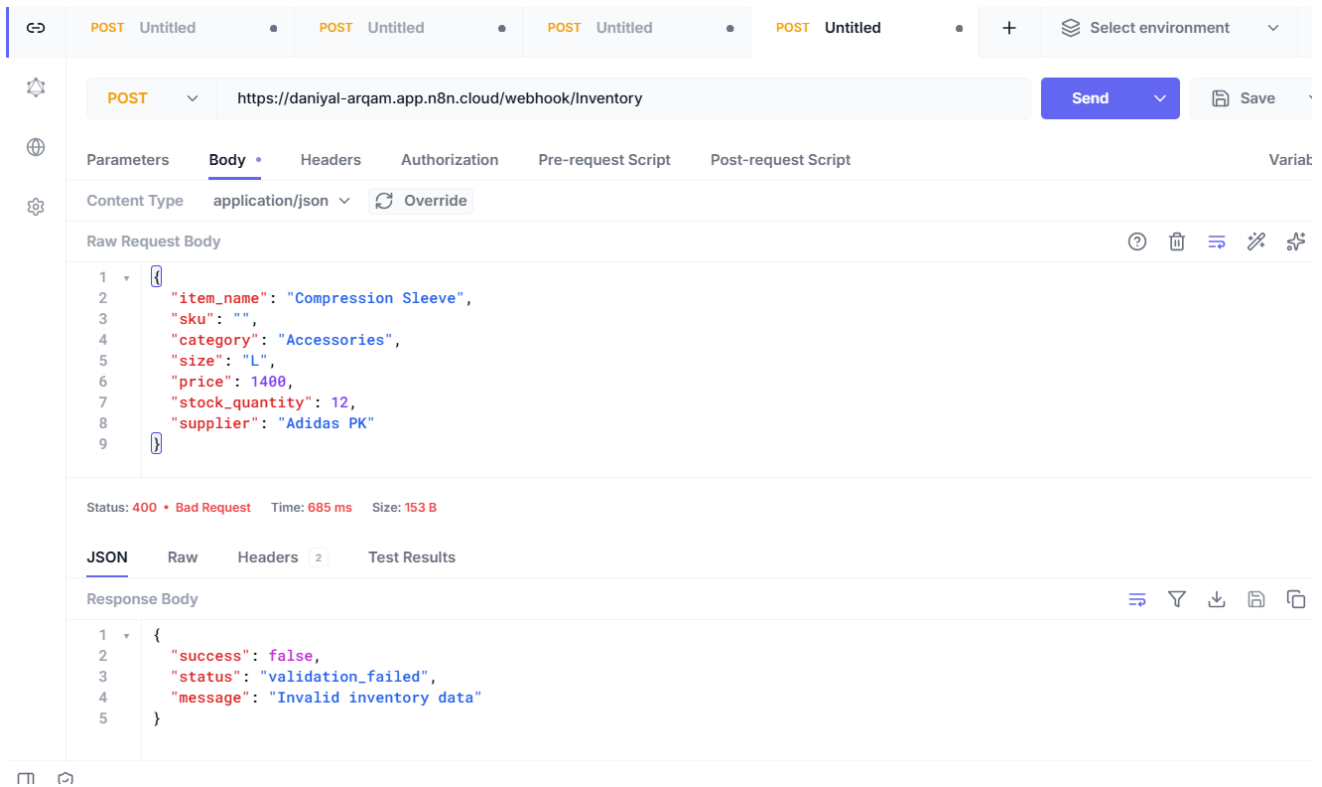


Figure 5: Missing SKU correctly failed validation.

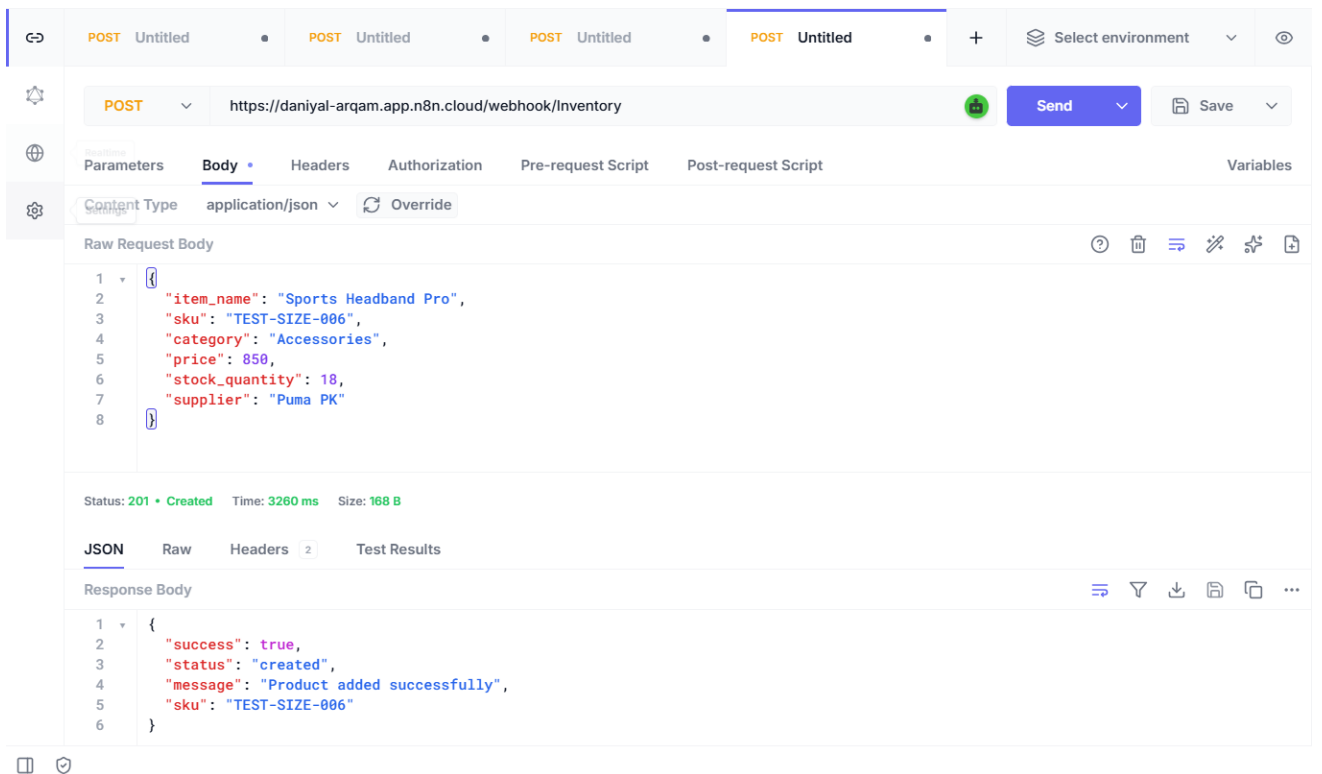


Figure 6: Optional size can be omitted and the request still succeeds.

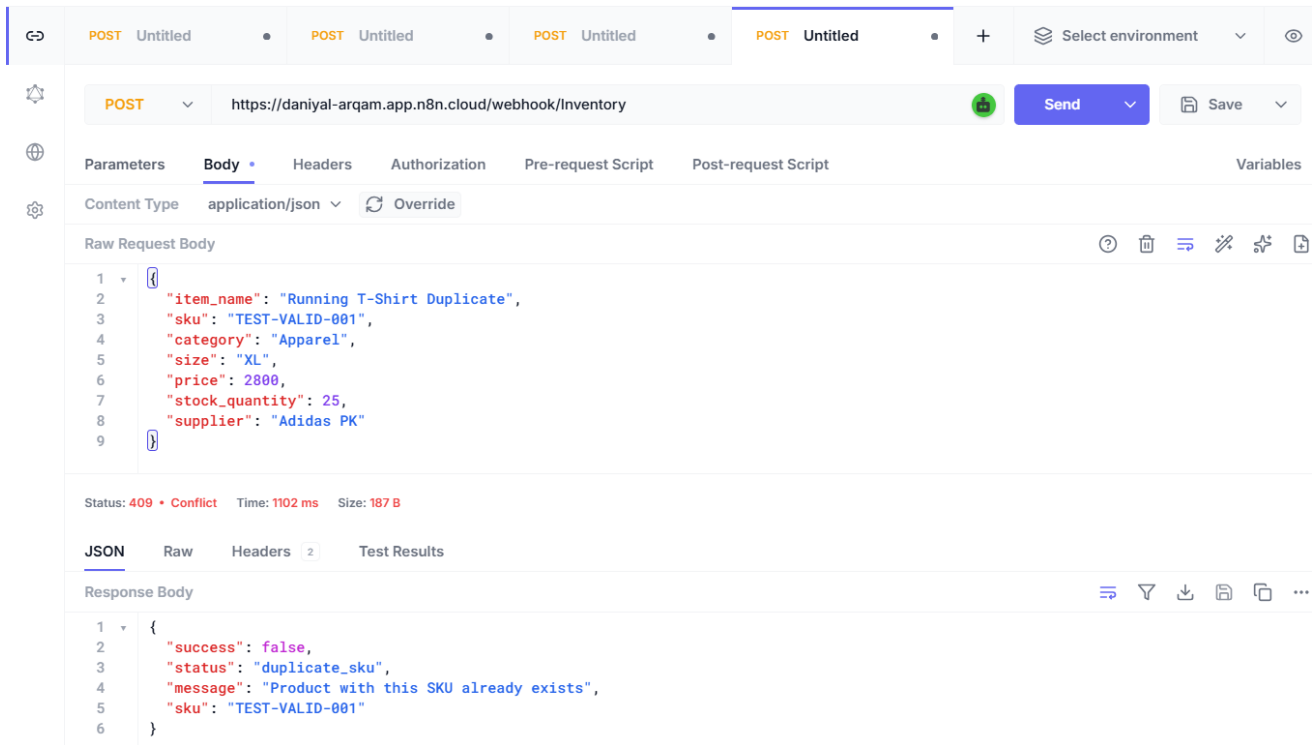


Figure 7: Duplicate SKU detection returned 409 Conflict.

5. AI Agent and Reorder Decision

The AI part is used only for low-stock products. The low-stock threshold is 5 or below. The AI Agent uses the Groq openai/gpt-oss-20b model and has a real Google Sheets tool connected to it. Its job is to classify the urgency of the stock situation, recommend a reorder quantity, give a short business reason, and save that recommendation in the Reorder_Alerts sheet.

For example, stock values from 0 to 2 are treated as CRITICAL and values from 3 to 5 as HIGH. The reorder quantity itself is recommended by the AI Agent rather than being hard-coded into the workflow.

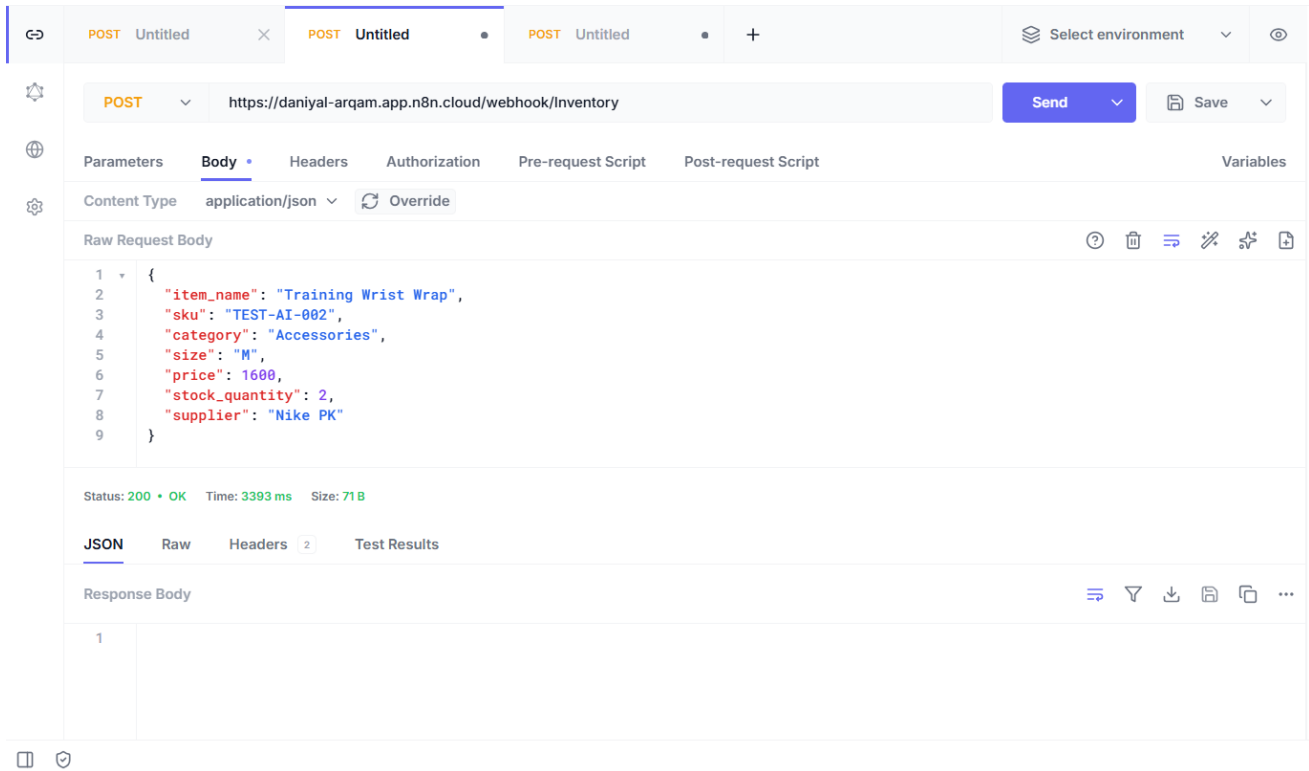


Figure 8: Low-stock test request sent through Hoppscotch. The AI result is evidenced by the saved reorder records below.

Sportsweat Inventory							
File Edit View Insert Format Data Tools Extensions Help							
100% 123 Default... 10 B I A							
D13	A	B	C	D	E	F	G
1	SKU	Item_Name	Current_Stock	Urgency	Recommended_Reorder_Qty	Supplier	Reason
2	WS-AI-004	Wrist Support Pro	1	CRITICAL	10	Nike PK	Stock is critically low, reorder 10 to prevent stockouts.
3	WS-AI-004	Wrist Support Pro	1	CRITICAL	9	Nike PK	Stock critically low, reorder to maintain inventory
4	GK-AI-003	Goalkeeper Gloves	2	CRITICAL	20	Puma PK	Stock at critical level, reorder 20 to maintain availability and meet demand.
5							
6							
7							
8							
9							
10							
11							
12							
13							
14							
15							
16							
17							
18							
19							
20							
21							
22							
23							

Figure 9: Reorder_Alerts sheet containing AI-generated urgency, reorder quantity and reason.

6. Stored Inventory Results

Valid inventory items are stored in the main Sheet1 tab. The sheet also shows the calculated stock status and timestamp. The missing-size test is visible with N/A, which confirms that the optional-field fallback works.

	A	B	C	D	E	F	G	H	I	J
	Item_Name	Sku	Category	Size	Price	Stock_Quantity	Supplier	Stock_status	Time_Stamp	
1	Running Shoes	NK-RS-001	Footwear	42	8500	25	Nike Pk	In Stock		
2	Dri-FIT Shirt	AD-TS-045	Apparel	L	3200	8	Adidas	In Stock		
3	Sports Cap	PU-CP-012	Accessories	One Size	1500	0	Puma	In Stock		
4	Running Shoes	NK-RS-001	Footwear	42	8500	25	Nike PK	In Stock	2026-08-27T10:08:08.679+05:00	
5	Football Boots	FB-NEW-001	Footwear	43	12500	15	Adidas PK	In Stock	2026-08-28T00:13:26.021+05:00	
6	Training Jacket	TJ-NEW-002	Apparel	L	6500	12	Adidas PK	In Stock	2026-08-28T00:37:52.938+05:00	
7	Training Jacket Pro	TJ-UNIQUE-827	Apparel	L	6500	12	Adidas PK	In Stock	2026-08-28T00:43:34.717+05:00	
8	Basketball Shoes	BS-TEST-9999	Footwear	44	11000	20	Nike PK	In Stock	2026-08-28T00:47:40.173+05:00	
9	Basketball Shoes	BS-UNIQUE-7777	Footwear	44	11000	20	Nike PK	In Stock	2026-08-28T00:52:00.383+05:00	
10	Gym Gloves	GG-NEW-901	Accessories	M	2200	18	Puma PK	In Stock	2026-08-28T01:02:54.109+05:00	
11	Gym Gloves Pro	GG-NEW-902	Accessories	M	2500	20	Puma PK	In Stock	2026-08-28T01:16:03.631+05:00	
12	Training Gloves	TG-AI-001	Accessories	M	2200	2	Adidas PK	Low Stock	2026-08-29T20:40:30.401+05:00	
13	Football Socks Pro	FS-AI-002	Accessories	L	1200	1	Nike PK	Low Stock	2026-08-29T20:54:03.679+05:00	
14	Goalkeeper Gloves	GK-AI-003	Accessories	L	3500	2	Puma PK	Low Stock	2026-08-29T21:07:10.645+05:00	
15	Wrist Support Pro	WS-AI-004	Accessories	M	1800	1	Nike PK	Low Stock	2026-08-29T21:14:11.720+05:00	
16	Sports Headband	EDGE-SIZE-001	Accessories	N/A	750	18	Puma PK	In Stock	2026-08-29T21:22:21.058+05:00	
17	Running T-Shirt	TEST-VALID-001	Apparel	L	2500	20	Adidas PK	In Stock	2026-08-29T21:54:18.277+05:00	
18	Sports Headband Pro	TEST-SIZE-006	Accessories	N/A	850	18	Puma PK	In Stock	2026-08-29T21:58:14.077+05:00	
19										
20										
21										
22										
23										

Figure 10: Main Google Sheet containing successfully validated inventory records.

7. Rule-Based Logic vs AI Decisions

I used normal workflow rules for decisions that should always give the same answer: required-field validation, price validation, negative-stock validation, the low-stock threshold, duplicate SKU detection, and HTTP response codes. I used AI for the reorder recommendation because that part benefits from a more flexible decision. This separation keeps the workflow reliable without using AI where simple rules are enough.

8. Response Handling

The API returns clear responses for the main outcomes. A newly created product returns 201 Created, invalid inventory data returns 400 Bad Request, and a duplicate SKU returns 409 Conflict. This makes it easy for another application or user to understand what happened after submitting inventory data.

9. Future Improvements

If I continued this project, I would add authentication for the webhook, category-specific low-stock thresholds, supplier email or API notifications, stock update and return operations, stronger SKU-format validation, and an audit log. I would also make reorder recommendations consider recent sales history so the AI can use demand patterns instead of current stock alone.

10. Conclusion

The final workflow handles both normal and problematic inventory inputs instead of only storing rows. It validates data, prevents duplicates, handles an optional field, stores structured inventory records, and gives the AI Agent a practical responsibility for low-stock reordering. The tests show that the main success and failure paths behave as expected.